

Large- R jets with radius parameter $R = 1.0$ associated with hadronically decaying top quarks are selected over jets originating from the fragmentation of other quarks or gluons by requiring that they contain several high- p_T objects and have a mass compatible with the top-quark mass. Standard object-selection criteria (defined in Table 3) are applied, with a *stricter* lepton p_T requirement and *tight* lepton identification criteria. The final event-selection criteria are:

- Single-electron or single-muon trigger satisfied;
- Exactly one lepton (electron or muon) with $p_T > 30$ GeV;
- E_T^{miss} larger than 20 GeV and $E_T^{\text{miss}} + M_T^W > 60$ GeV;
- At least one small- R jet close to the lepton, i.e. with $\Delta R(\text{small-}R \text{ jet, lepton}) < 2.0$;
- Exactly one large- R jet, passing the simplified top-tagging requirements [57]: a mass larger than 100 GeV and N-subjettiness ratio $\tau_{32} < 0.75$;
- This *top-tagged* large- R jet must be well separated from the lepton, $\Delta\phi(\text{large-}R \text{ jet, lepton}) > 1.0$, and from the small- R jet associated with the lepton, $\Delta R(\text{large-}R \text{ jet, small-}R \text{ jet}) > 1.5$;
- At least one b -tagged (MV2c10 @ 70% WP) small- R jet that fulfils the following requirements: it is either inside the top-tagged large- R jet, $\Delta R(\text{large-}R \text{ jet, } b\text{-tagged jet}) < 1.0$, or it is the small- R jet associated with the lepton.

The dominant source of background comes from $t\bar{t}$ production, studied in Section 3.3, and the shapes of all background distributions are modelled with MC simulation. No multijet background estimation is attempted and the contributing effects of fake or non-prompt leptons are considered to be negligible.

An observable that roughly approximates the mass of the $t\bar{t}$ system is constructed [56] by summing the four-momentum of the top-tagged large- R jet, the four-momenta of the charged lepton and the b -tagged small- R jet associated with the lepton. For simplicity, the neutrino momentum is not added. Reconstructing the full $t\bar{t}$ system and studying its properties is possible, but introduces a slight challenge due to the reconstruction of the z component of the neutrino's momentum [58], which may be seen as an interesting educational challenge.

The distributions of the large- R jet mass before and after the top-tagging requirements, top-tagged large- R jet pseudorapidity, azimuthal angle, N-subjettiness, transverse momentum before and after a systematic uncertainty variation and the approximate mass of the $t\bar{t}$ system are shown in Figure 11. Good agreement within statistical uncertainties is found between data and MC prediction. The benchmark 1 TeV Z' model is overlaid, clearly showing the different kinematic properties of this particular BSM prediction.

3.12 Two-photon final state: the case of SM Higgs boson production in the $H \rightarrow \gamma\gamma$ decay channel

The $H \rightarrow \gamma\gamma$ decay mode provides a very clear and distinctive signature of two isolated and highly energetic photons, and is one of the main channels studied at the LHC. Despite the small branching ratio, $(2.27 \pm 0.07) \times 10^{-3}$ for $m_H = 125$ GeV [43], a reasonably large signal yield can be obtained thanks to the high photon reconstruction and identification efficiency at the ATLAS experiment. Furthermore, due to the excellent photon energy resolution of the ATLAS calorimeter, the signal manifests itself as a narrow peak in the diphoton invariant mass ($m_{\gamma\gamma}$) spectrum on top of a smoothly falling irreducible background from QCD production of two photons.

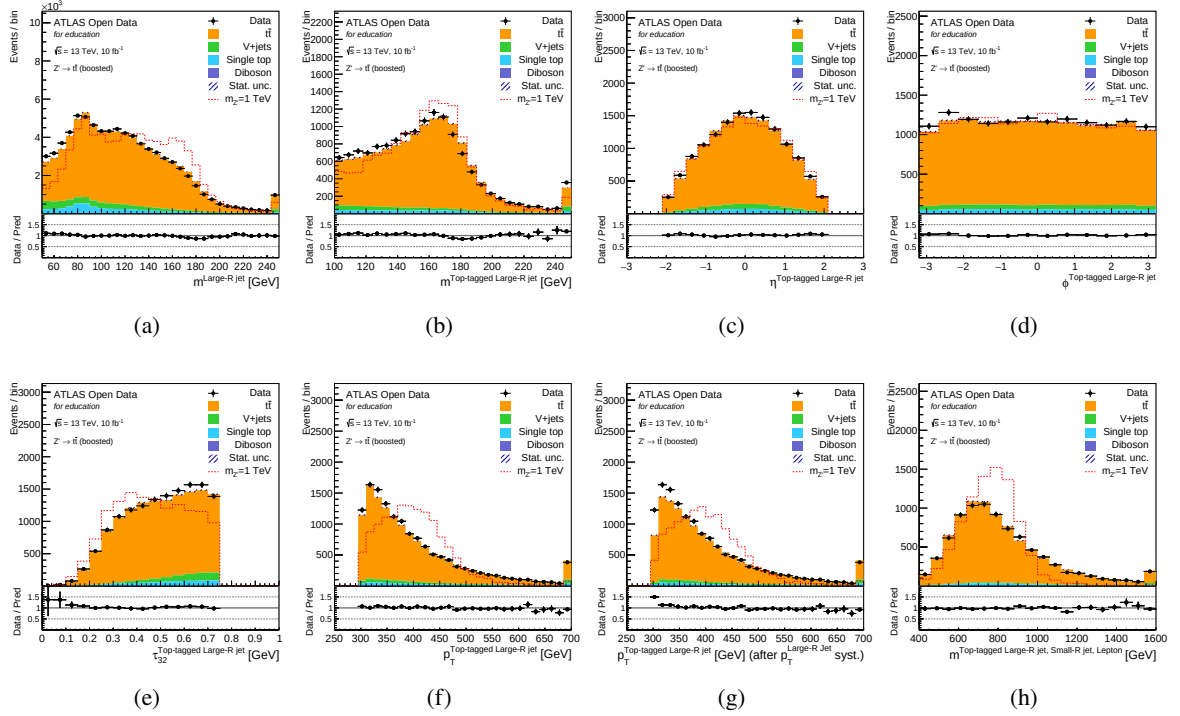


Figure 11: Comparison between data and MC prediction for several distributions in the $Z' \rightarrow t\bar{t}$ single-lepton boosted selection. Distributions of the large- R jet mass (a) before and (b) after the top-tagging requirements, top-tagged large- R jet (c) pseudorapidity, (d) azimuthal angle, (e) N-subjettiness, (f) transverse momentum, (g) transverse momentum after a systematic uncertainty variation and (h) approximate mass of the $t\bar{t}$ system (neutrino momentum not added). The points represent experimental data. The filled histograms show the prediction from different MC simulations. The contributions are stacked. The benchmark 1 TeV Z' signal prediction is overlaid as dotted red curve. The statistical uncertainty is represented by the error bars on the data points and the hashed area on the MC prediction. The last bin in all figures contains the overflow. The lower panels in each figure show the ratio of the data points to the stacked histogram.

The analysis presented is focused on implementing the selection criteria for $H \rightarrow \gamma\gamma$ event candidates and is loosely based on the measurement of Higgs boson in the diphoton decay channel with $\sqrt{s} = 13$ TeV data collected with the ATLAS detector [59]. Standard object-selection criteria (see Table 3) are applied. The final event-selection criteria are:

- Diphoton trigger is satisfied;
- Exactly two photons with $E_T > 35$ and 25 GeV, respectively;
- Leading and subleading photon candidates are respectively required to have $E_T/m_{\gamma\gamma} > 0.35$ and 0.25;
- Diphoton invariant mass $m_{\gamma\gamma}$ between 105 GeV and 160 GeV.

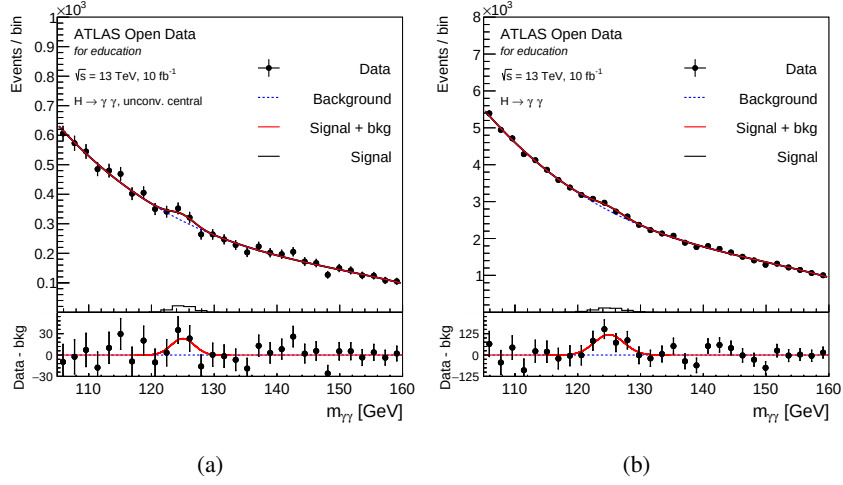


Figure 12: Diphoton invariant-mass spectrum as observed in the selected diphoton events, with (a) two unconverted central ($|\eta| < 0.75$) photons and (b) inclusively (for both unconverted and converted photons). The solid red curve shows the fitted signal-plus-background model when the Higgs boson mass is constrained to be 125 GeV. The background component of the fit is shown with the dotted blue curve. The signal component of the fit is shown with a solid black curve. The bottom plot shows the residuals between the data and the background component of the fitted model.

The background is estimated from data, without the use of MC simulation, by fitting the diphoton invariant-mass distribution in a range ($105 < m_{\gamma\gamma} < 160$ GeV) with a third-order polynomial function with free shape and normalisation parameters. Signal MC simulations of the five main Higgs-boson production mechanisms (ggF , VBF , WH , ZH , $t\bar{t}H$) are used to model the shape of the invariant mass of the signal, modelled as a Gaussian function.

The distributions of the invariant mass, $m_{\gamma\gamma}$, of the selected diphoton events are shown in Figure 12 in the case of events with two unconverted central ($|\eta| < 0.75$) photons and inclusively (for both unconverted and converted photons). In the inclusive case, a total of 76394 events in the data are selected and approximately 400 signal events for $m_H = 125$ GeV are expected in the diphoton invariant mass range between 105 GeV and 160 GeV. The result of a fit including a signal component fixed to $m_H = 125$ GeV and a background component described by a third-order polynomial is superimposed. An excess, corresponding to the production of the SM Higgs boson, is visible in data near an invariant mass value of 125 GeV.